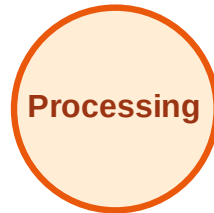


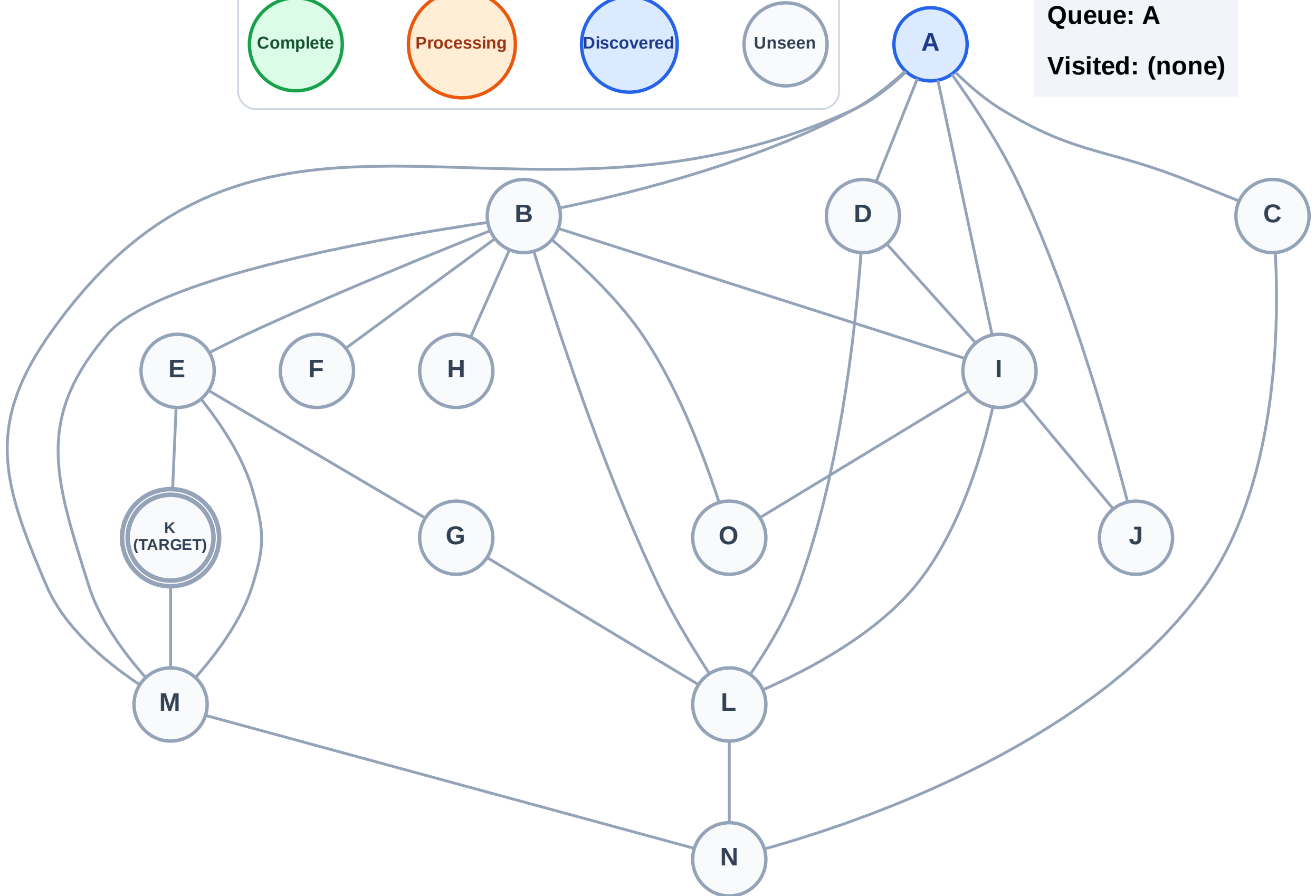
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



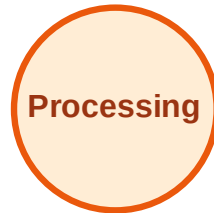
Queue: A
Visited: (none)



BFS Traversal

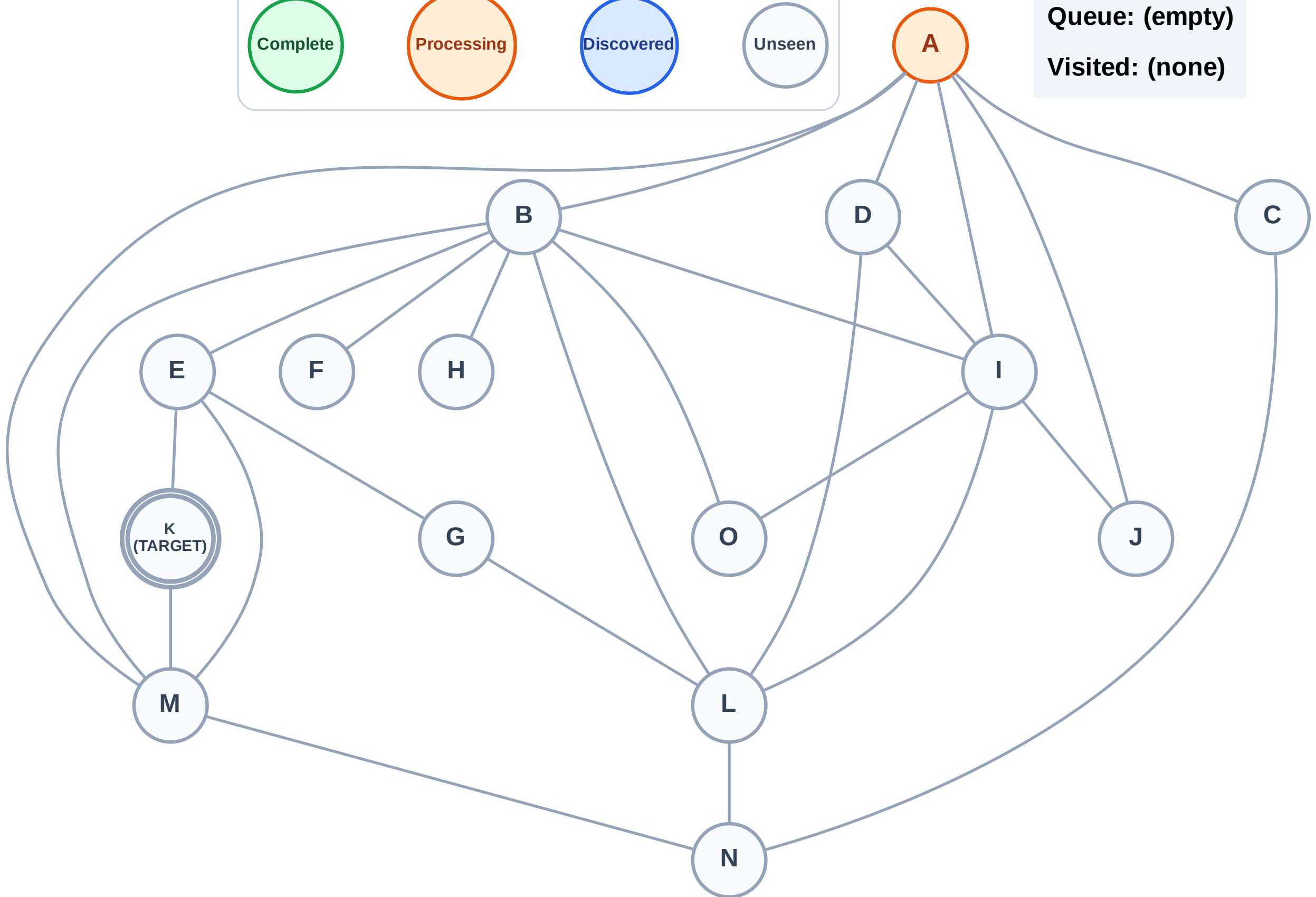
Step 2: Process node A

Legend



Queue: (empty)

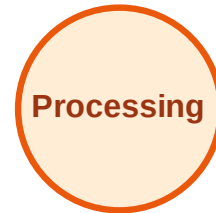
Visited: (none)



BFS Traversal

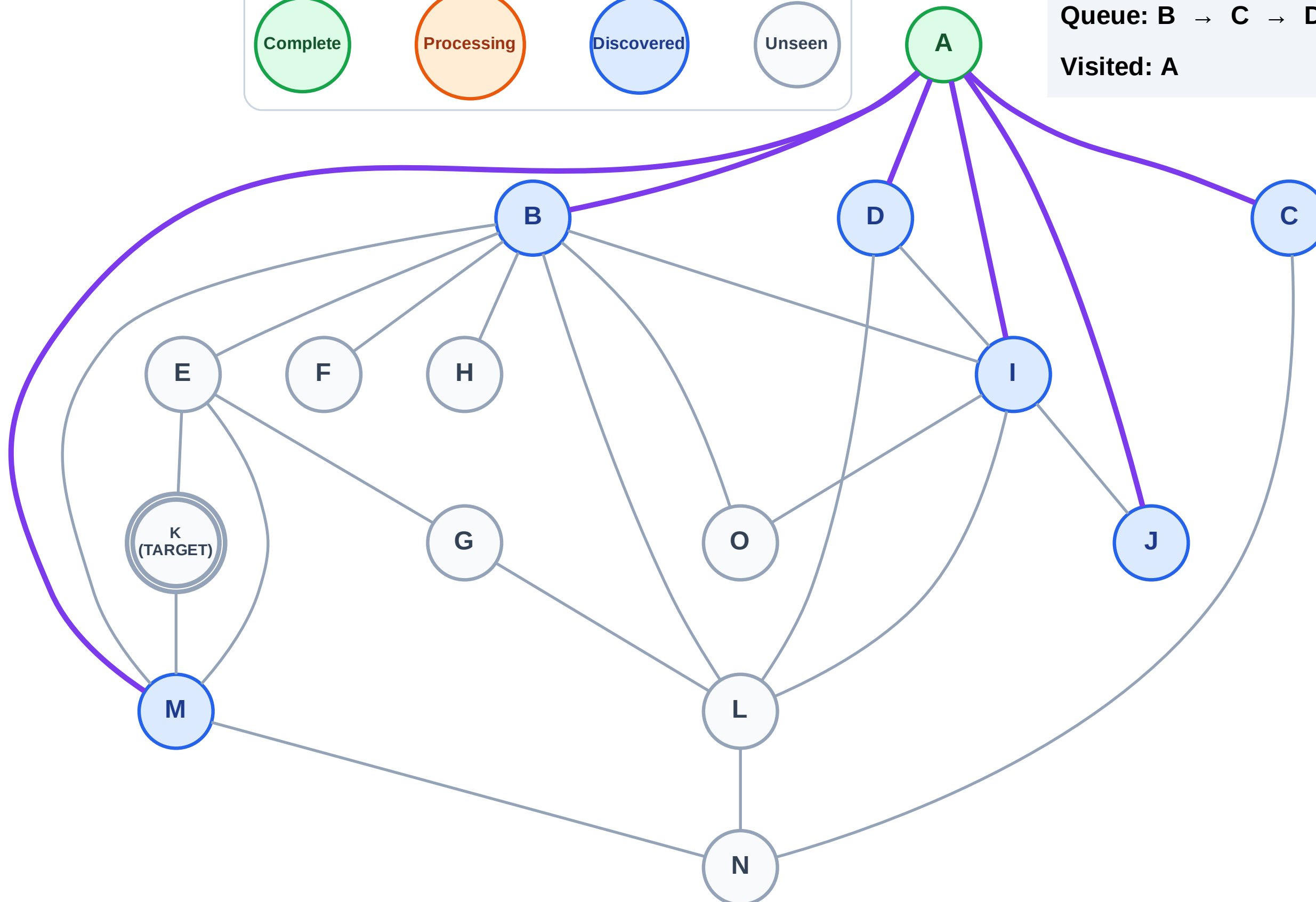
Step 3: Discover B, C, D, I, J, M from A

Legend



Queue: B → C → D → I → J → M

Visited: A



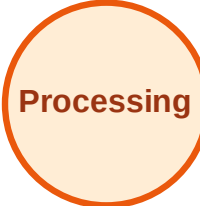
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



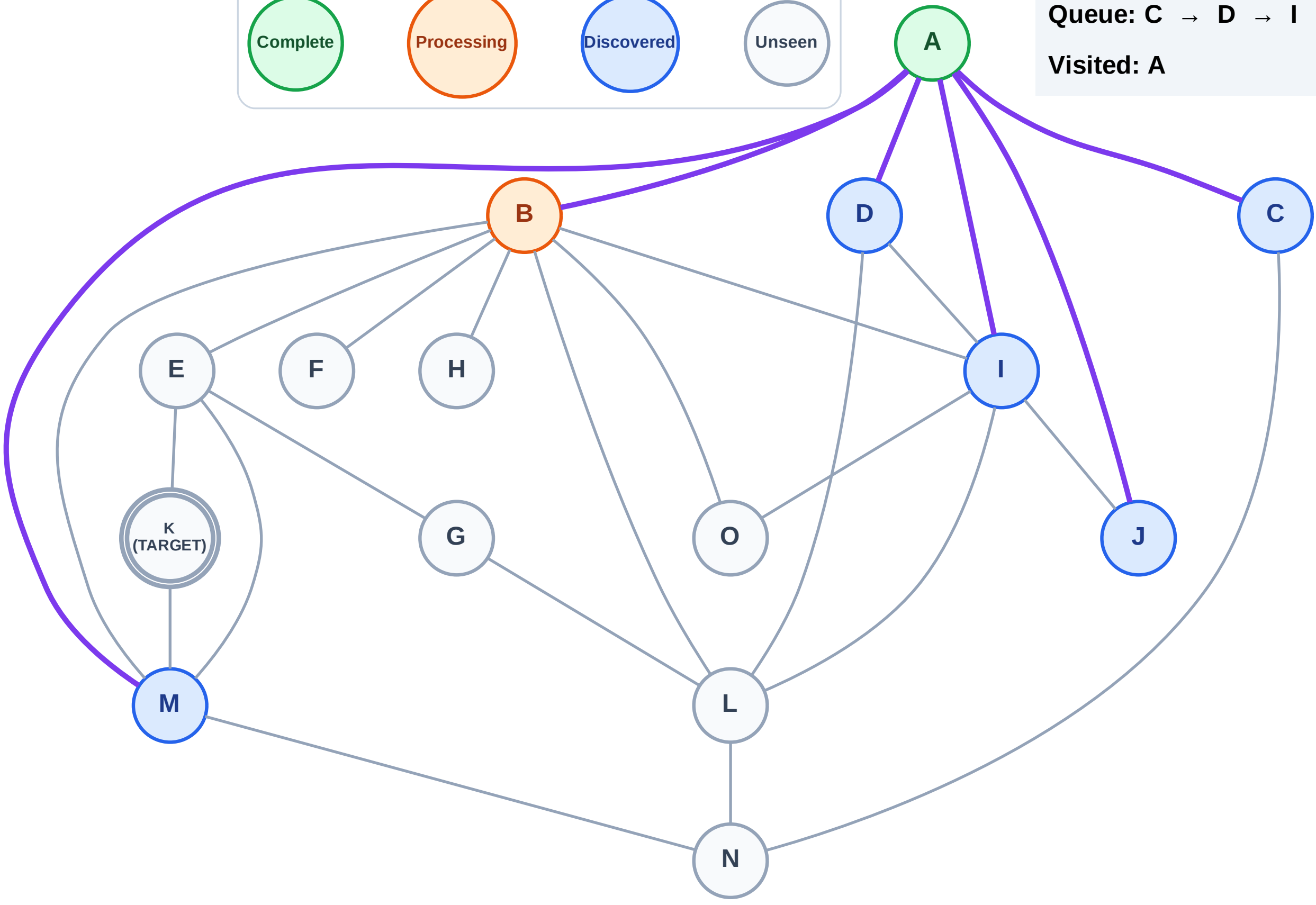
Discovered



Unseen

Queue: C → D → I → J → M

Visited: A



BFS Traversal

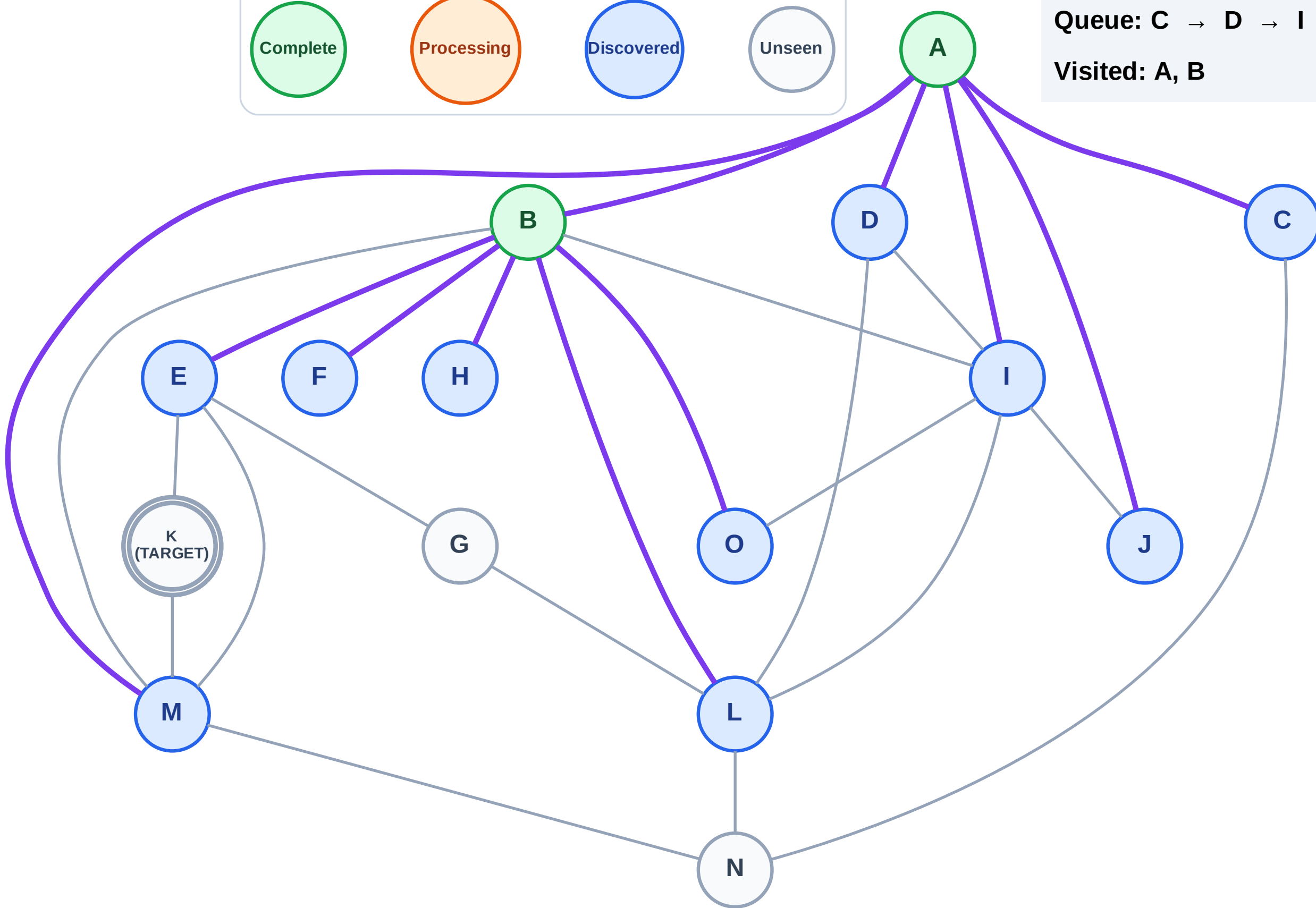
Step 5: Discover E, F, H, L, O from B

Legend



Queue: C → D → I → J → M → E → F → H → L → O

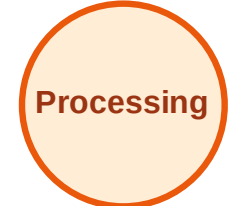
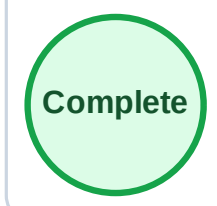
Visited: A, B



BFS Traversal

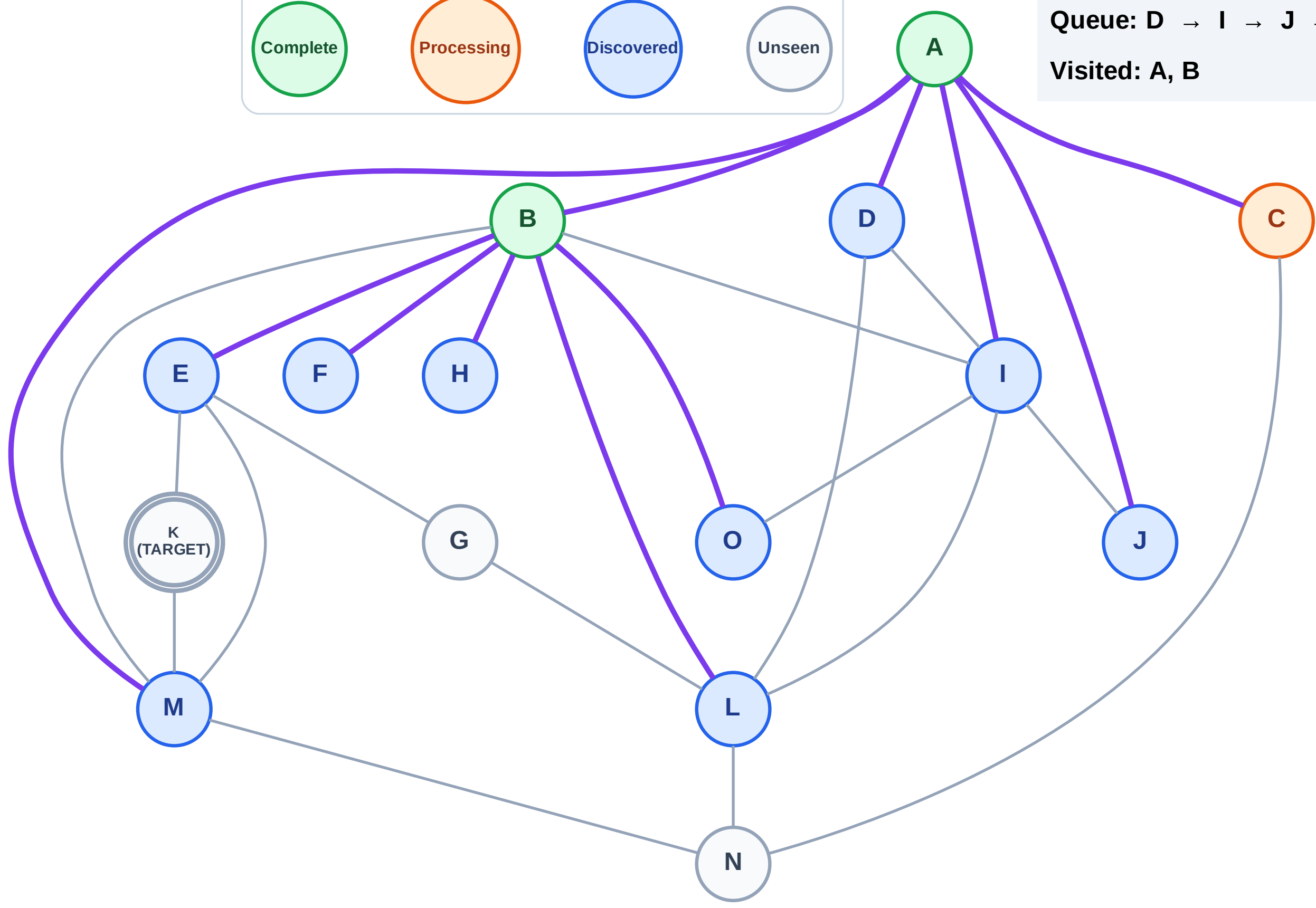
Step 6: Process node C

Legend



Queue: D → I → J → M → E → F → H → L → O

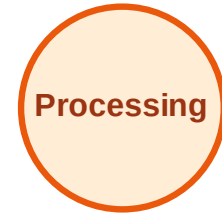
Visited: A, B



BFS Traversal

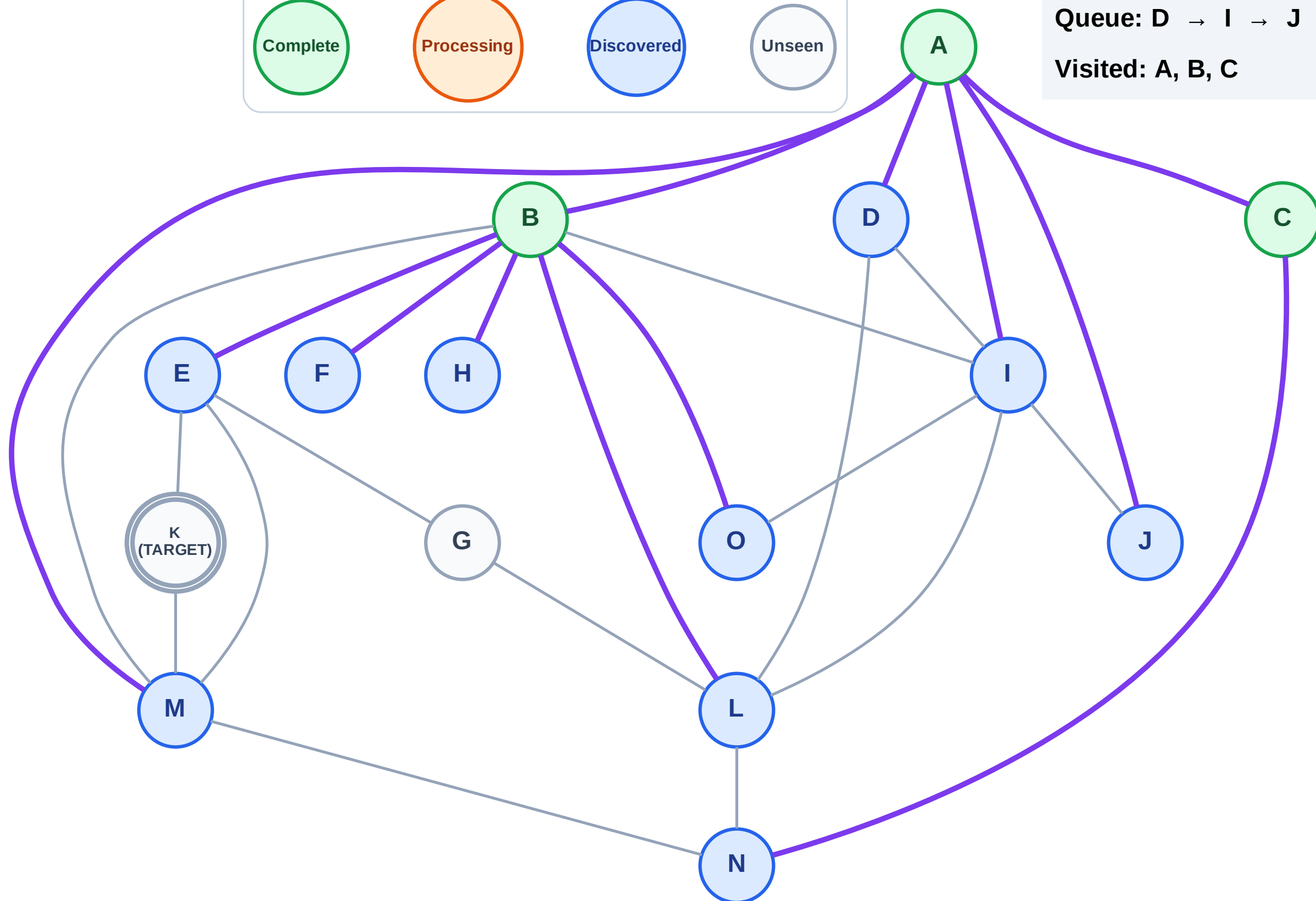
Step 7: Discover N from C

Legend



Queue: D → I → J → M → E → F → H → L → O → N

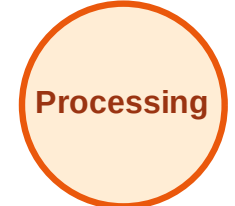
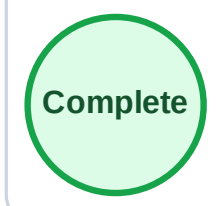
Visited: A, B, C



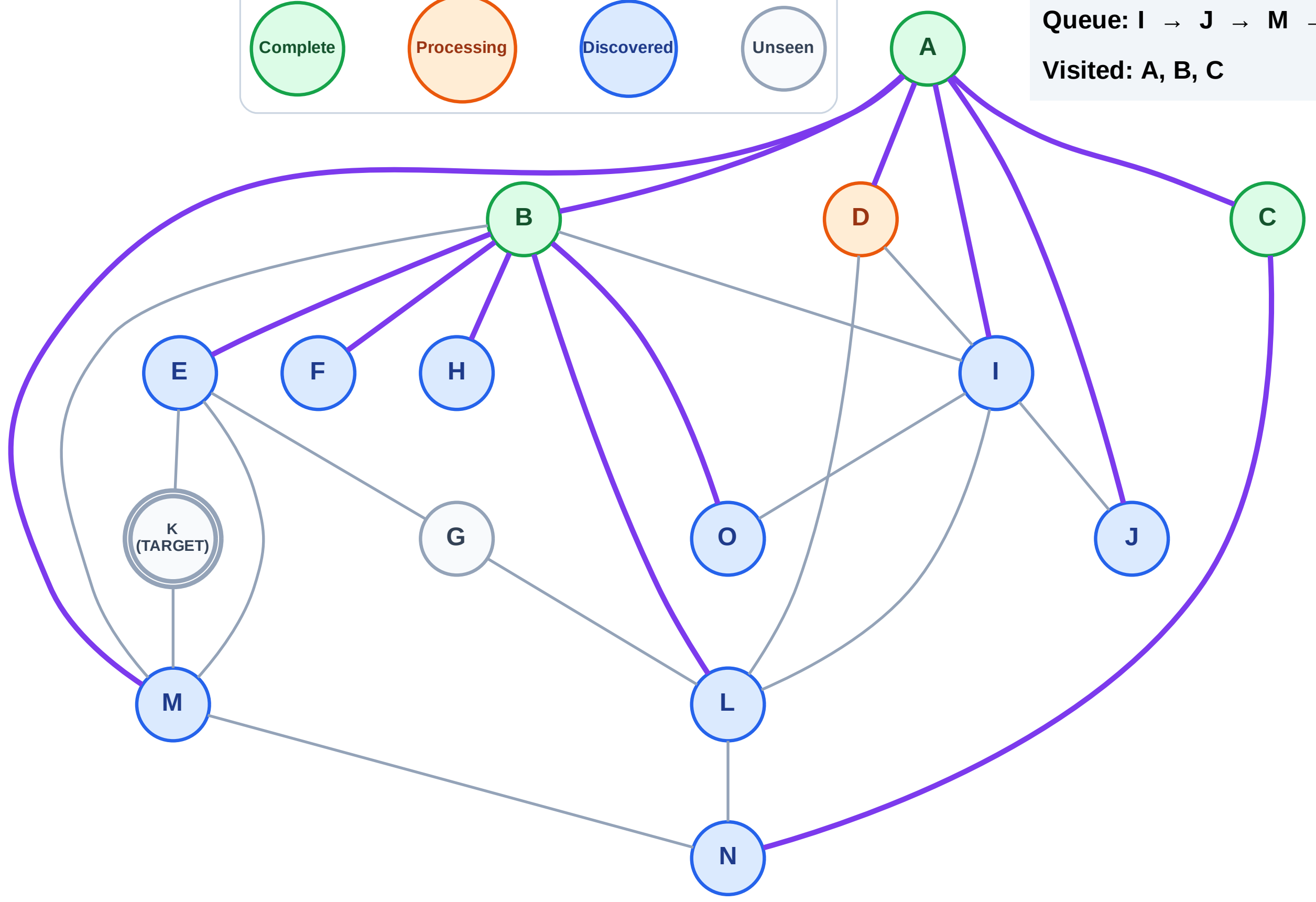
BFS Traversal

Step 8: Process node D

Legend



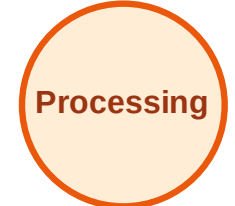
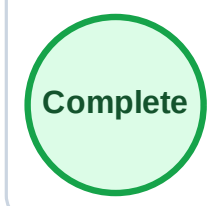
Queue: I → J → M → E → F → H → L → O → N
Visited: A, B, C



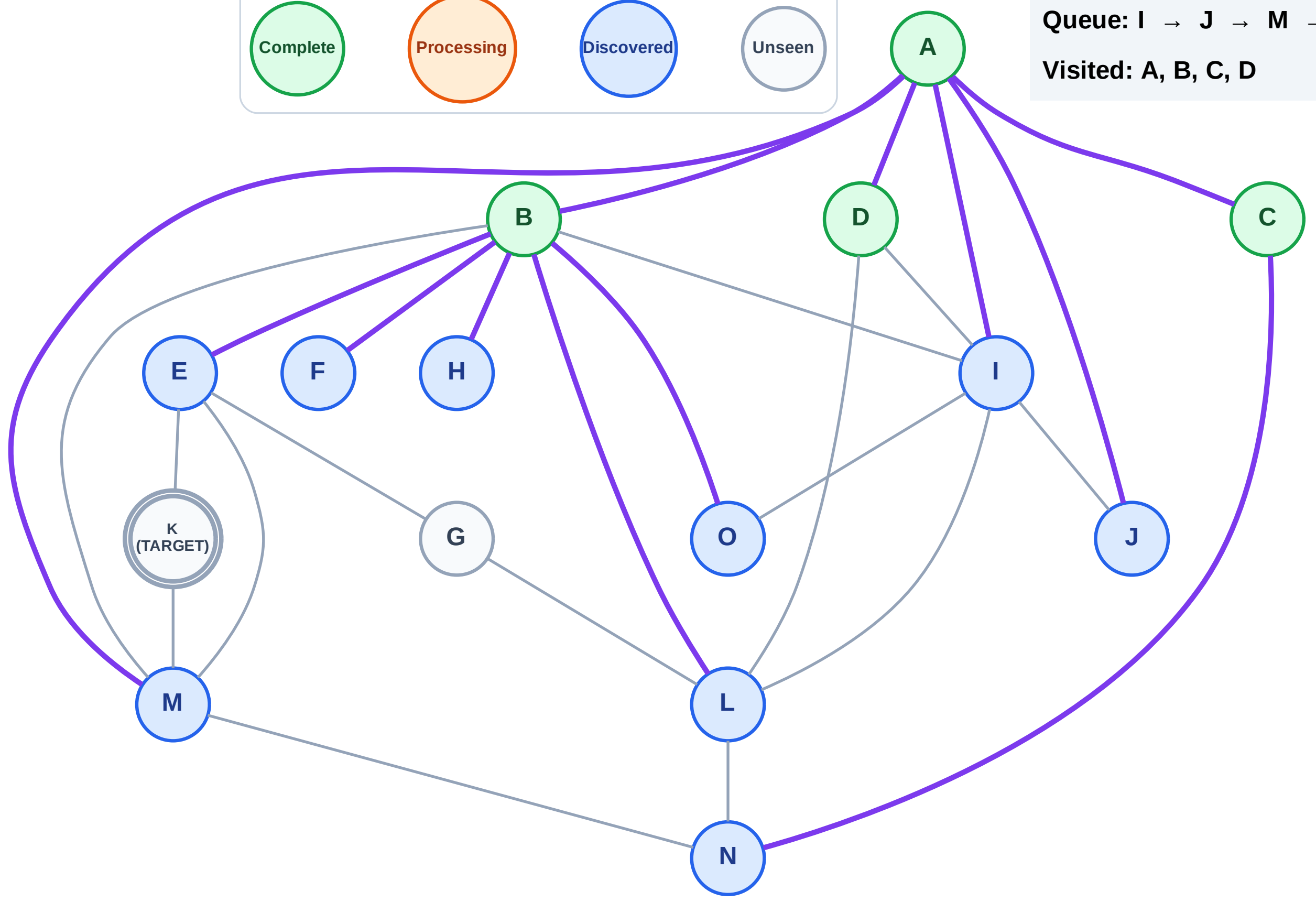
BFS Traversal

Step 9: No new neighbors from D

Legend



Queue: I → J → M → E → F → H → L → O → N
Visited: A, B, C, D



BFS Traversal

Step 10: Process node I

Legend

Complete

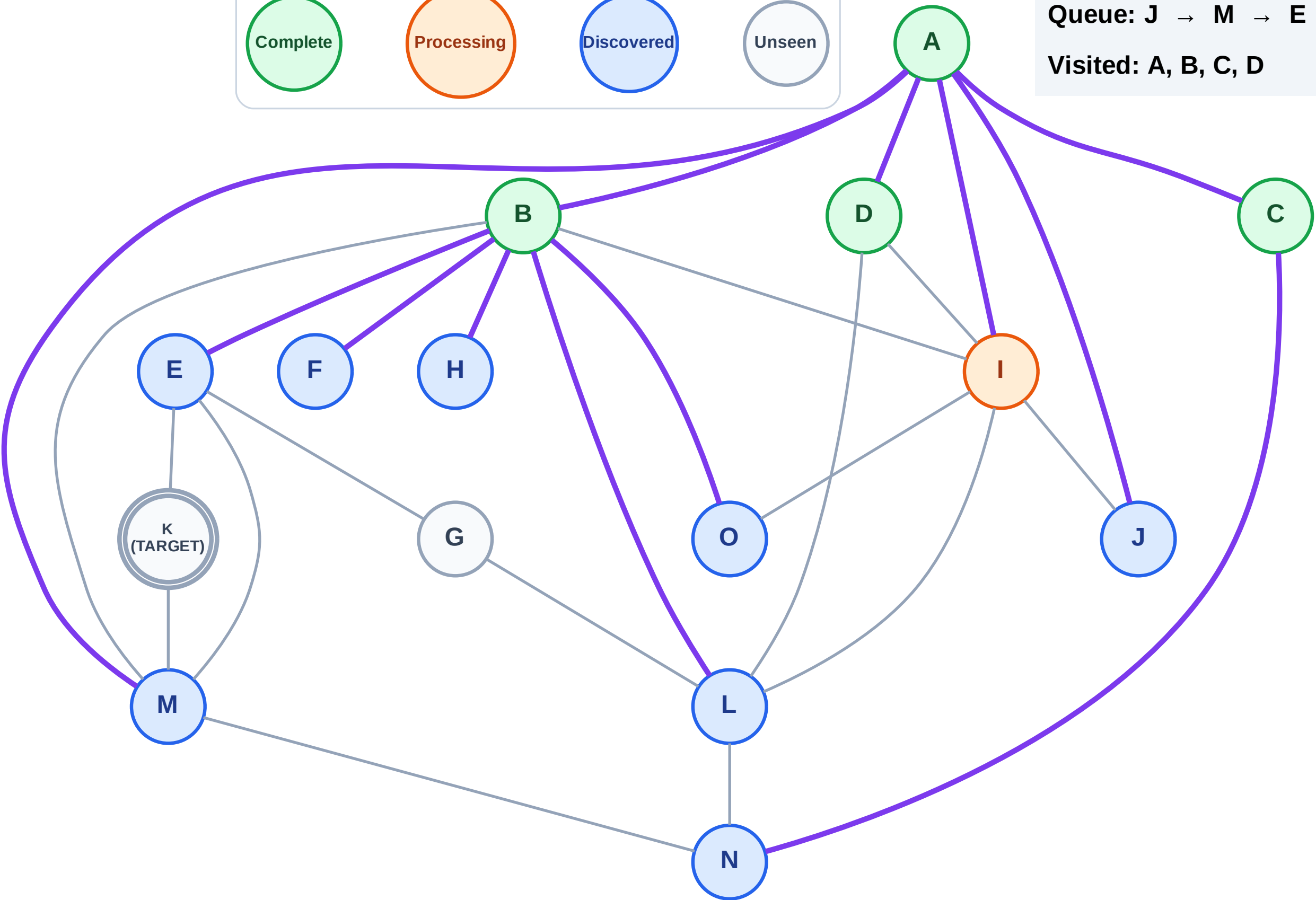
Processing

Discovered

Unseen

Queue: J → M → E → F → H → L → O → N

Visited: A, B, C, D



BFS Traversal

Step 11: No new neighbors from I

Legend

Complete

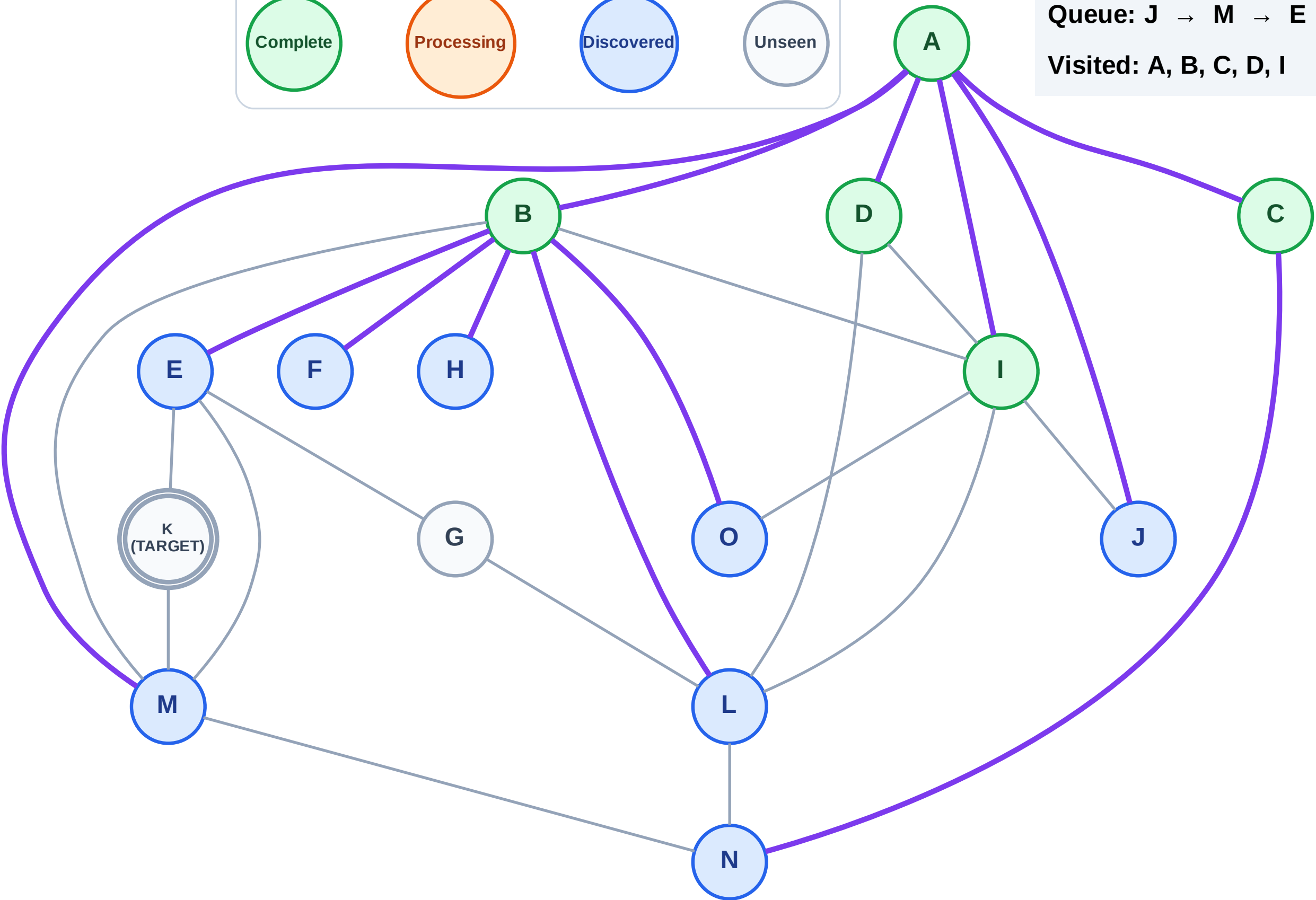
Processing

Discovered

Unseen

Queue: J → M → E → F → H → L → O → N

Visited: A, B, C, D, I



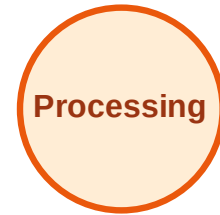
BFS Traversal

Step 12: Process node J

Legend



Complete



Processing



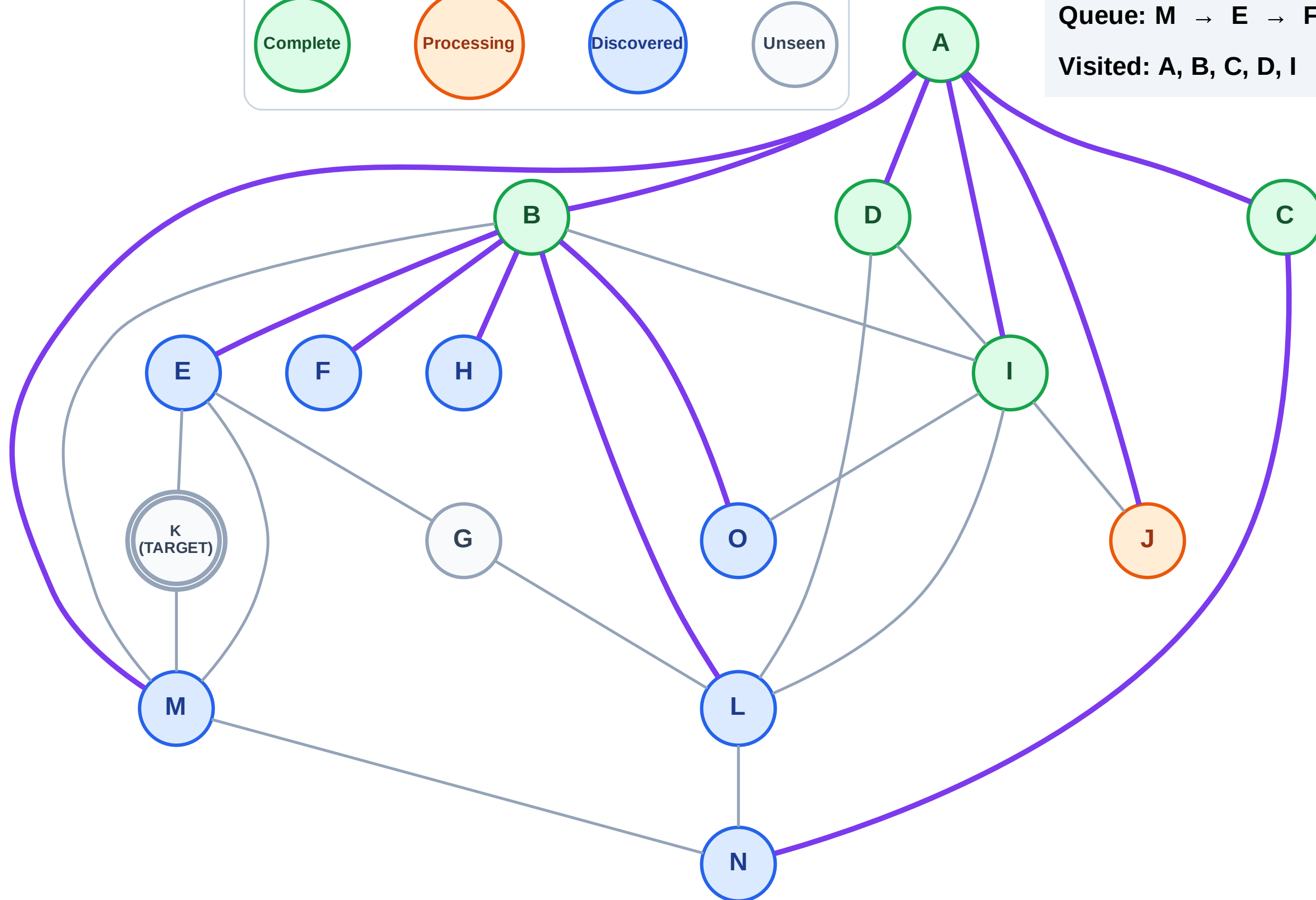
Discovered



Unseen

Queue: M → E → F → H → L → O → N

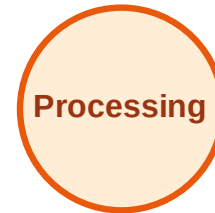
Visited: A, B, C, D, I



BFS Traversal

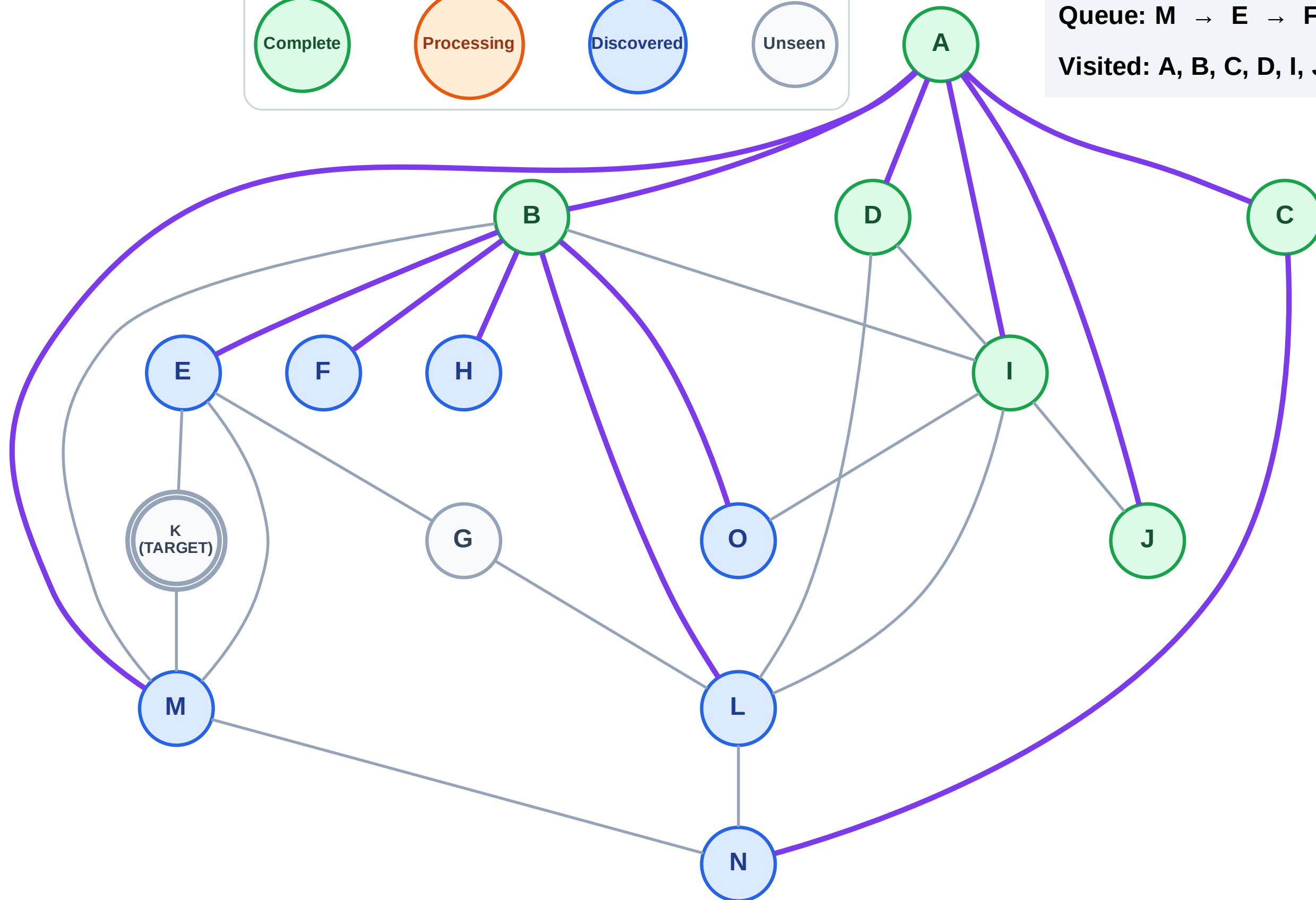
Step 13: No new neighbors from J

Legend



Queue: M → E → F → H → L → O → N

Visited: A, B, C, D, I, J



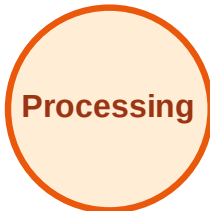
BFS Traversal

Step 14: Process node M

Legend



Complete



Processing



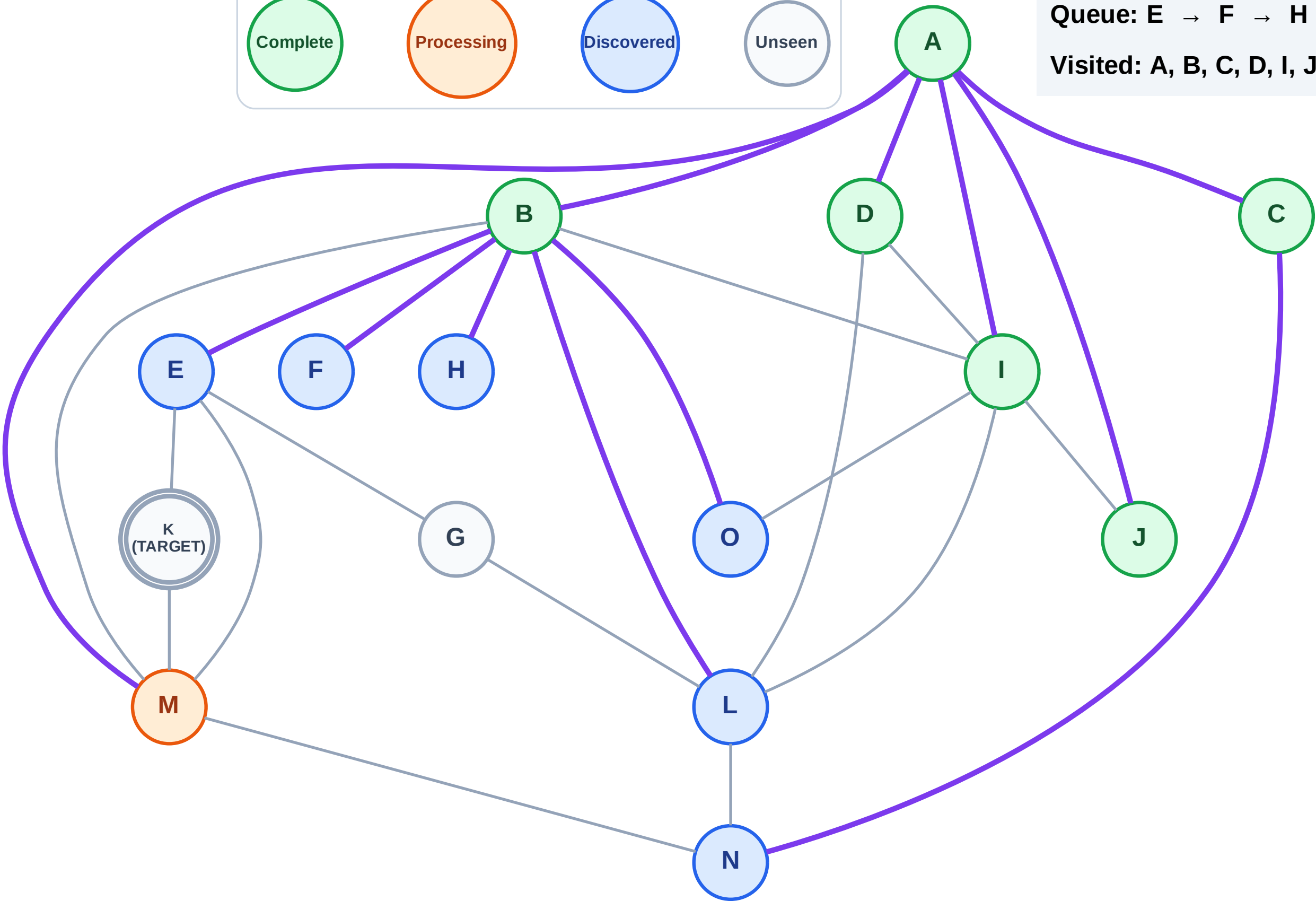
Discovered



Unseen

Queue: E → F → H → L → O → N

Visited: A, B, C, D, I, J



BFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → M → K

Legend

Complete

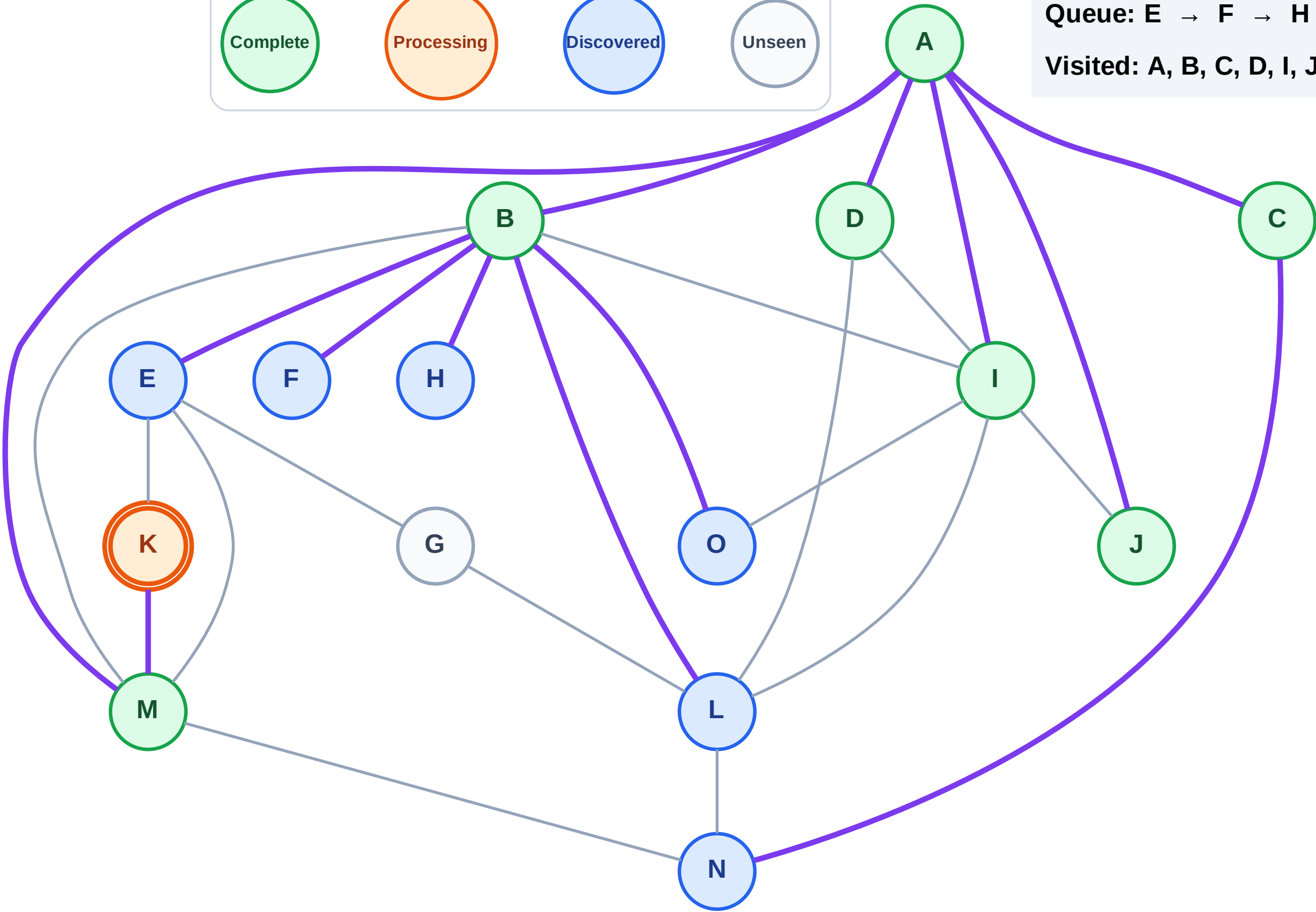
Processing

Discovered

Unseen

Queue: E → F → H → L → O → N

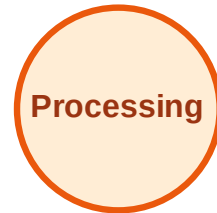
Visited: A, B, C, D, I, J, K, M



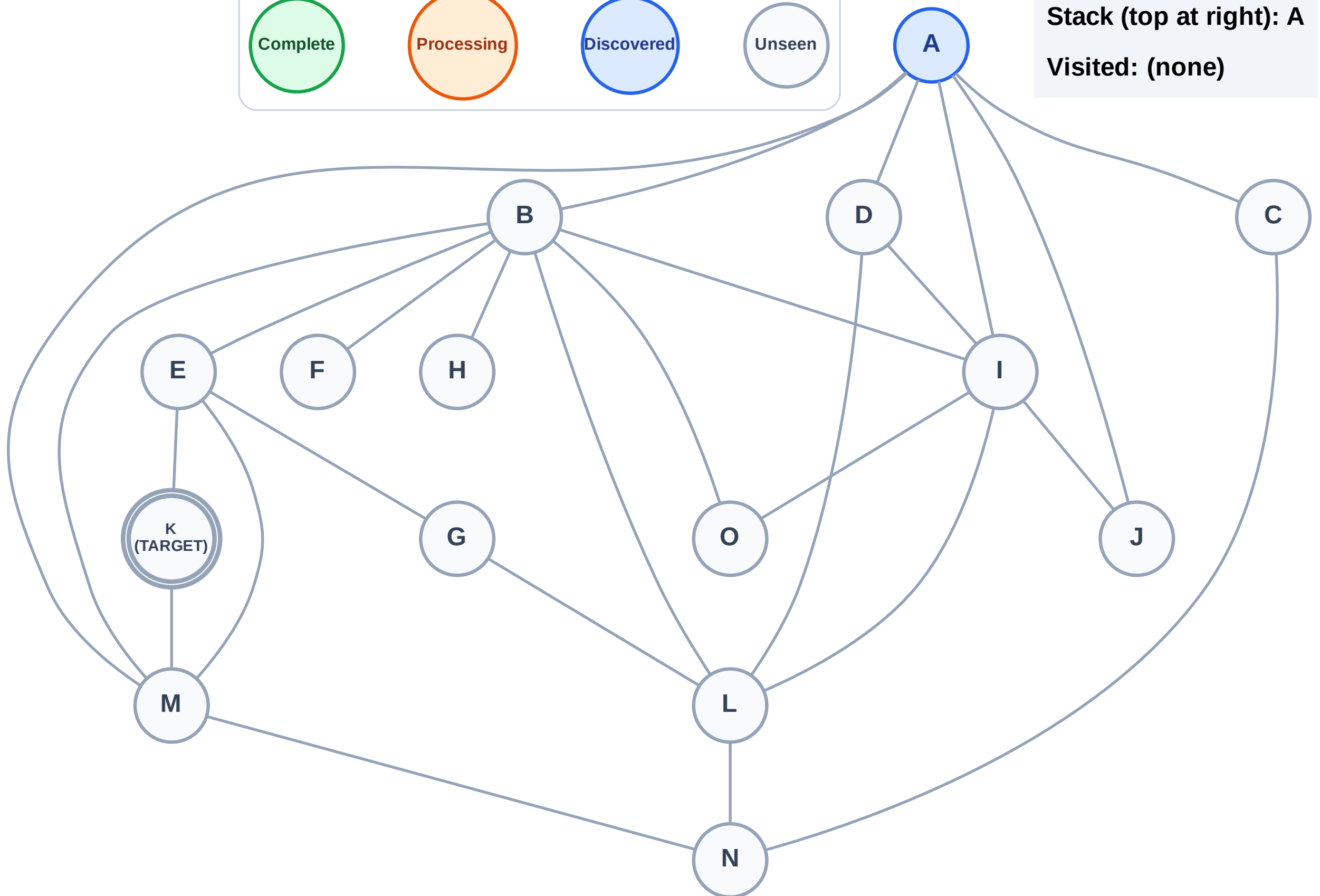
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A
Visited: (none)



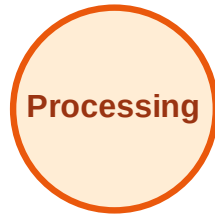
DFS Traversal

Step 2: Process node A

Legend



Complete



Processing



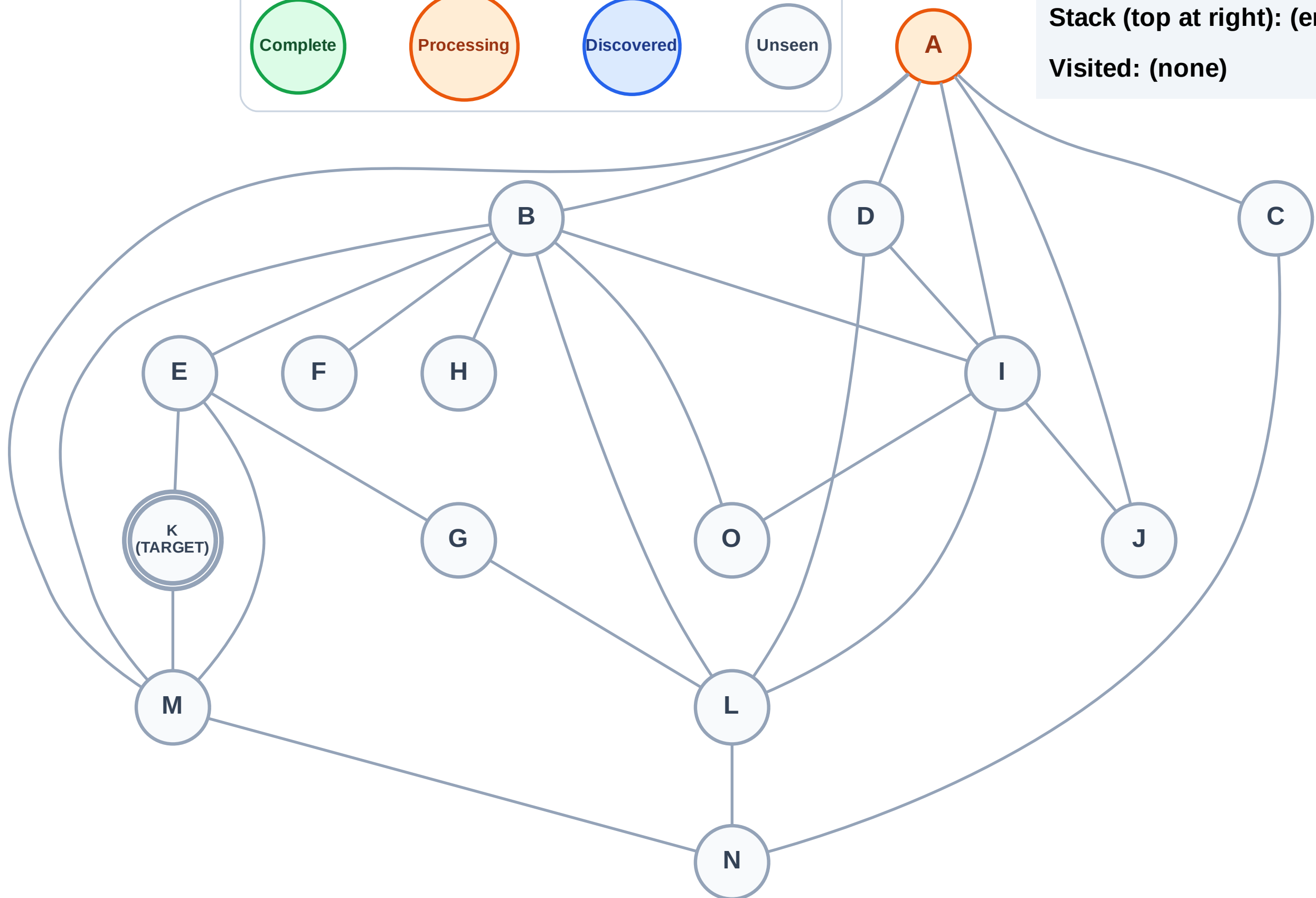
Discovered



Unseen

Stack (top at right): (empty)

Visited: (none)



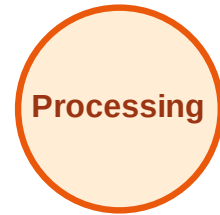
DFS Traversal

Step 3: Discover M, J, I, D, C, B from A

Legend



Complete



Processing



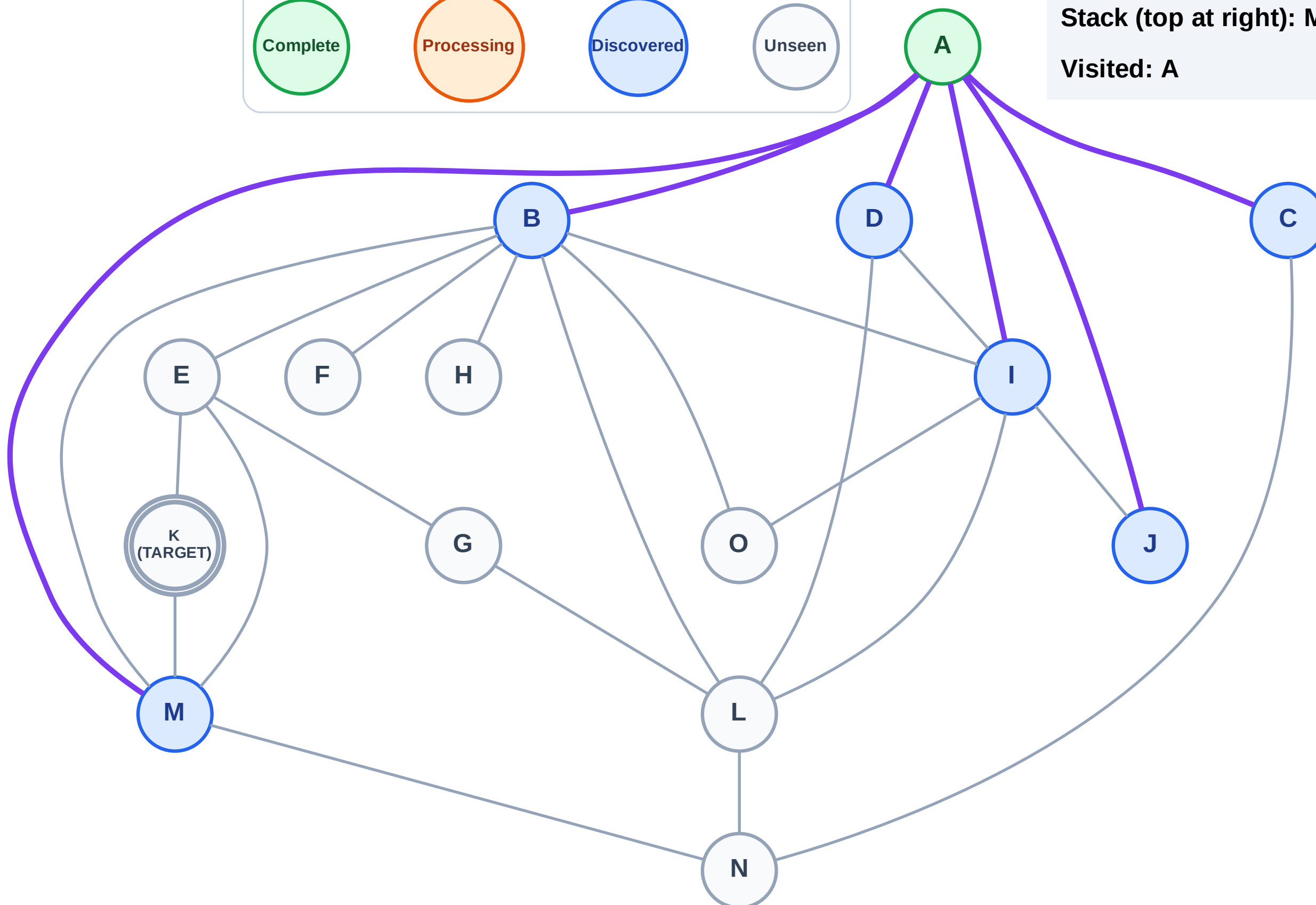
Discovered



Unseen

Stack (top at right): M | J | I | D | C | B

Visited: A



DFS Traversal

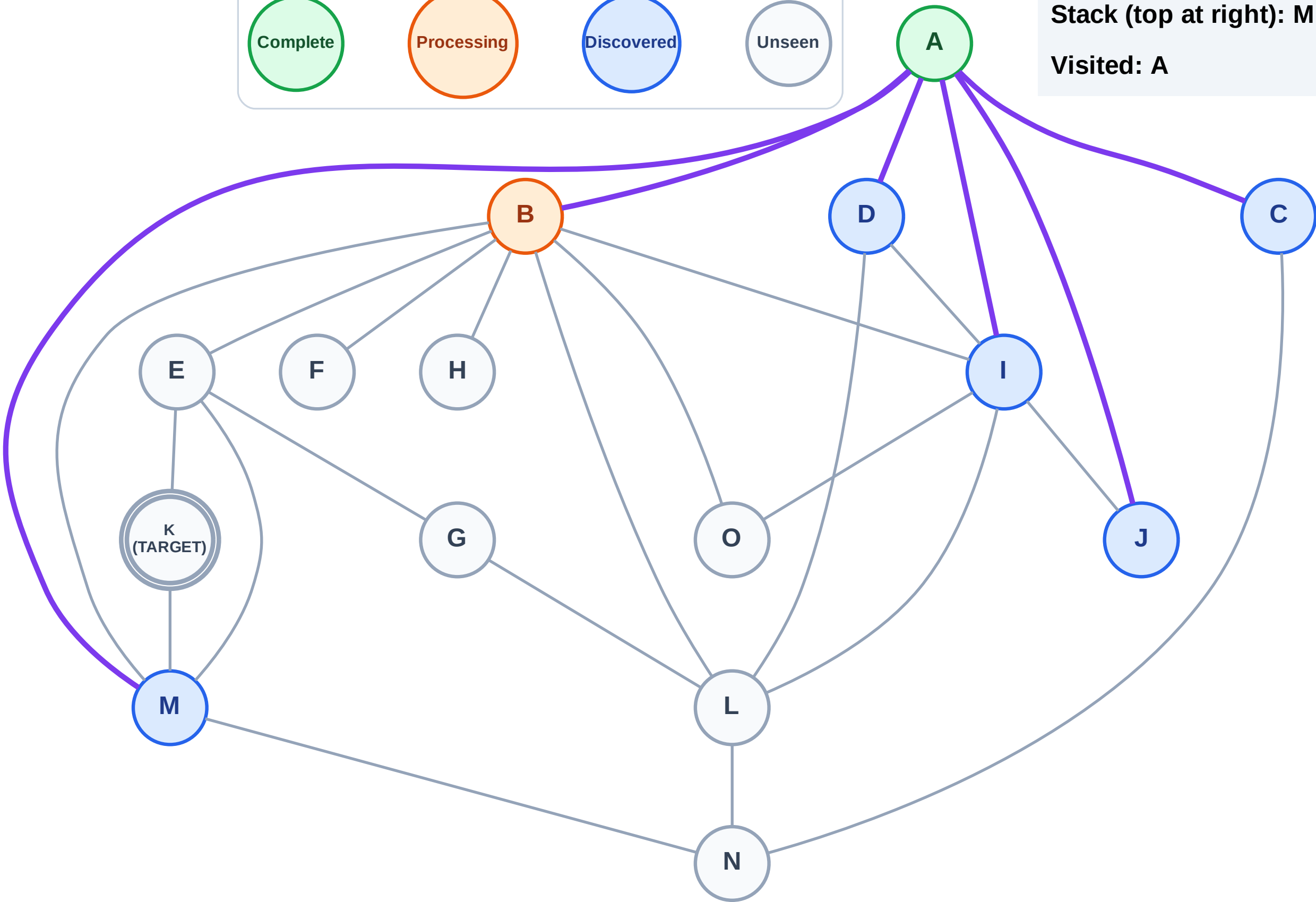
Step 4: Process node B

Legend



Stack (top at right): M | J | I | D | C

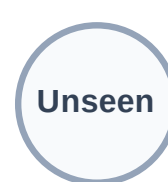
Visited: A



DFS Traversal

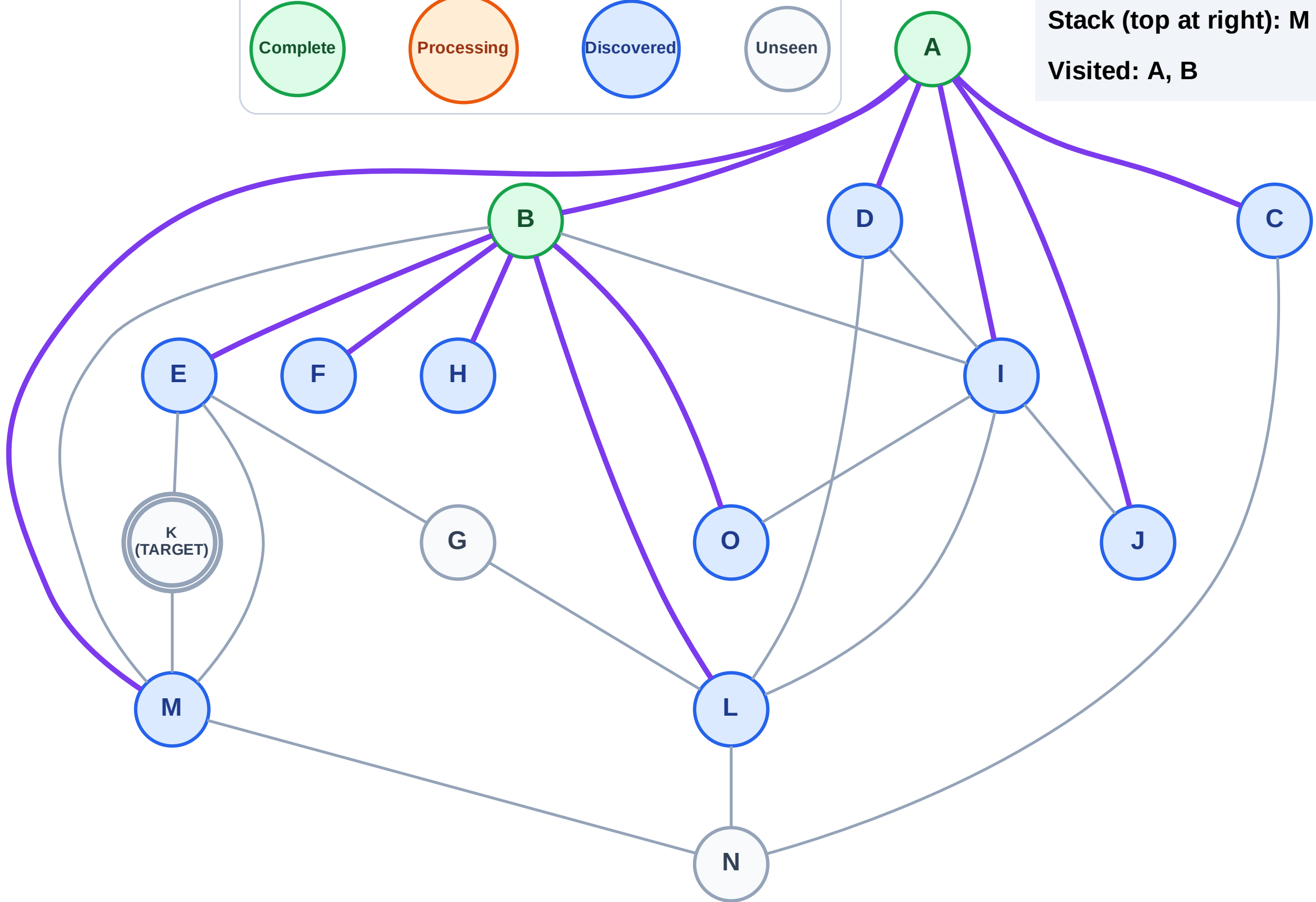
Step 5: Discover O, L, H, F, E from B

Legend



Stack (top at right): M | J | I | D | C | O | L | H | F | E

Visited: A, B



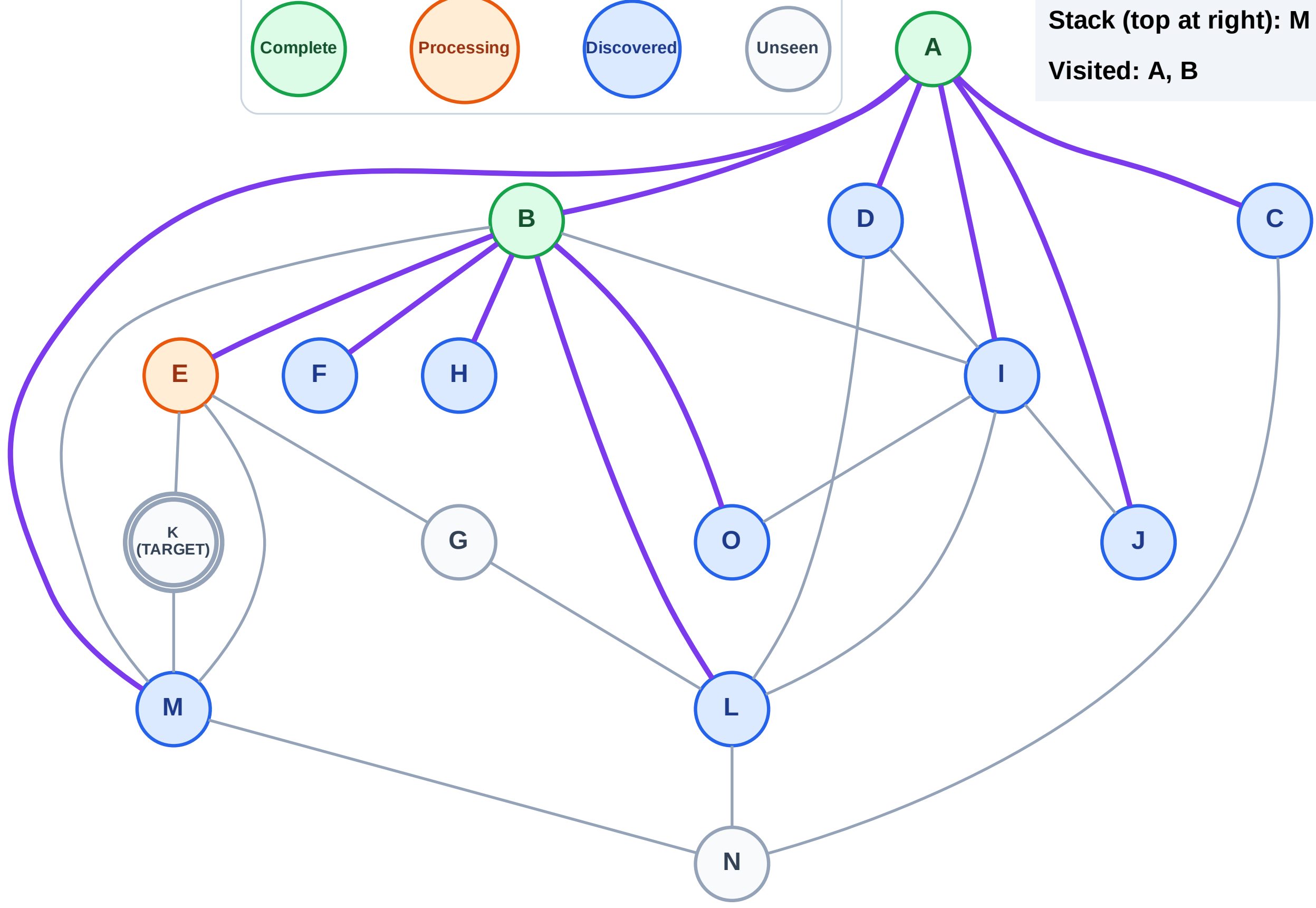
DFS Traversal

Step 6: Process node E

Legend



Stack (top at right): M | J | I | D | C | O | L | H | F
Visited: A, B



DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → E → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | J | I | D | C | O | L | H | F

Visited: A, B, E, K

